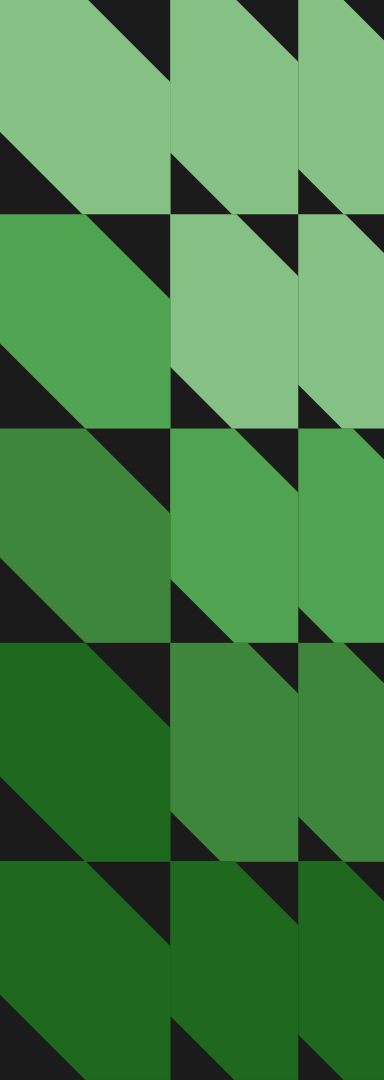




The Big Mac Language By: Zhang Chen and Isaac Miller

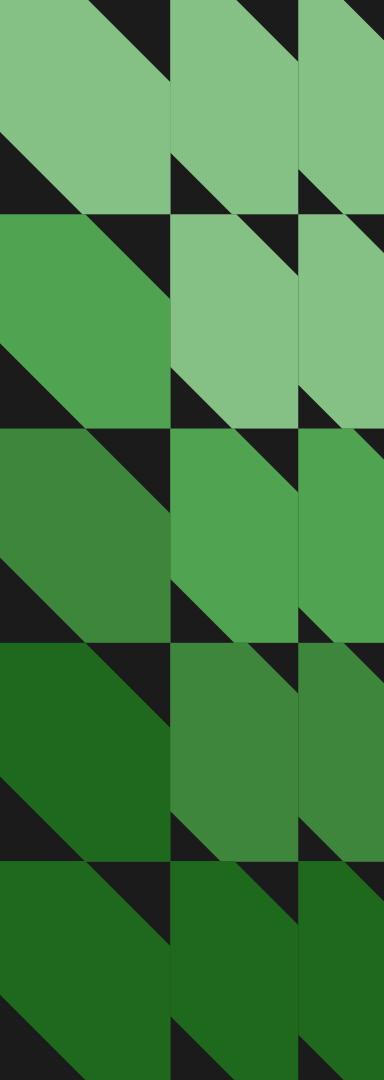


Added Features

We added two features to our language:

- Dot product operation:
 - Allow user to input two equal length vector and return a single scalar value
- Set operations:
 - Allows the user to input and perform set operations such as inserting into a set, showing the union between two sets, showing the intersection between two sets, showing the difference between two sets, showing that an element is a member of a set, and showing the size of a given set.

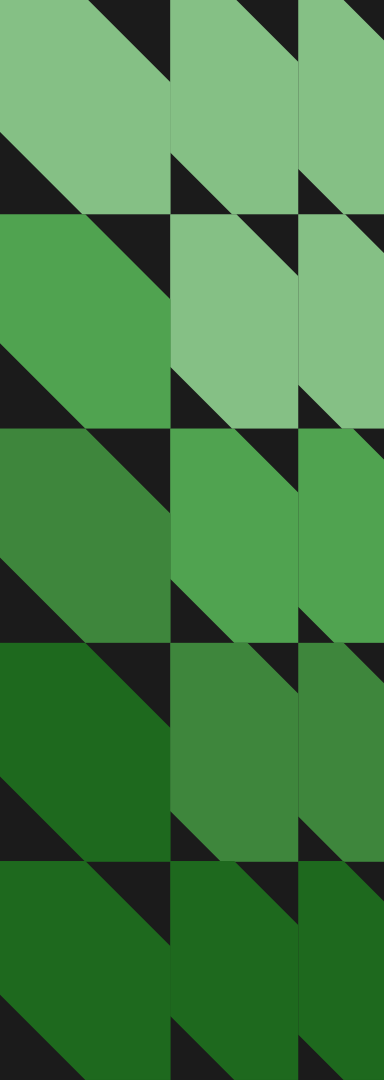
(Our language has nothing to do with Big Macs from McDonalds)



The Importance of these features

By adding the dot product feature, the language now can handle a vector-like list and being able to do dot product operation without hassle.

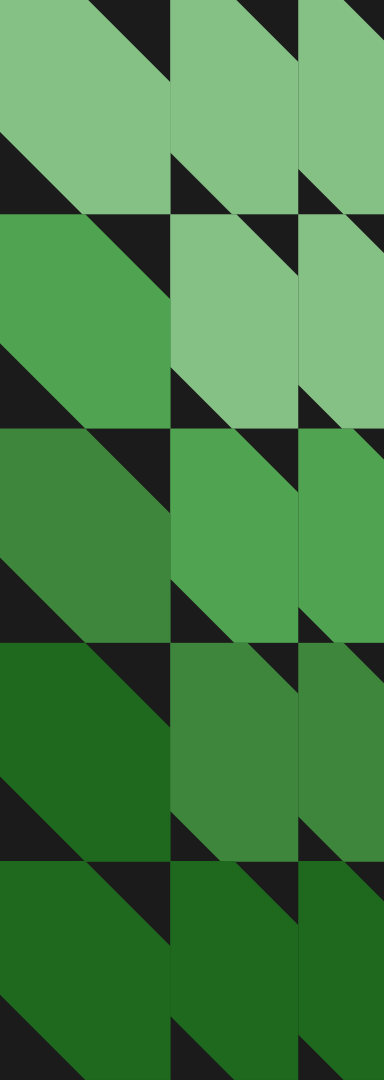
Using Set operations, the language now supports the use of the Set data structure, which allows the user to store a collection of distinct elements and perform operations with those elements.



Why we added them

Dot product operations is added to the language to show off the vector/list implementation and what the list/vector can achieve.

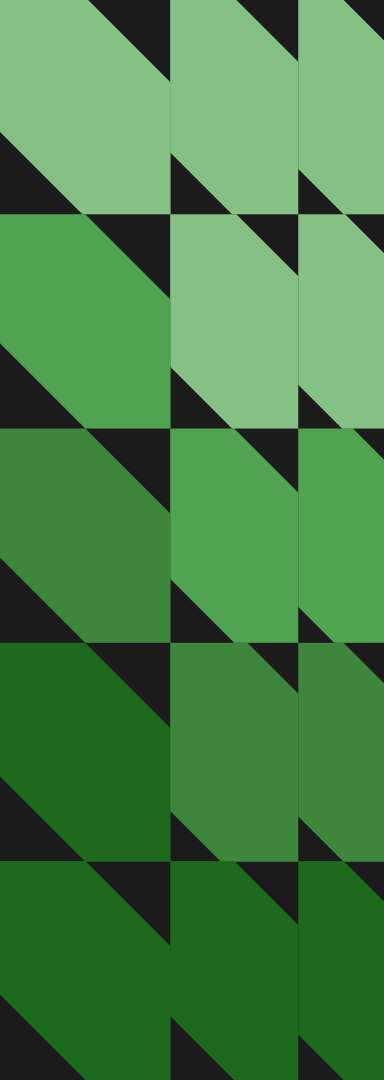
Set operations were added to the language to help demonstrate how sets, which are a very important data structure, can be implemented and support the various operations that come with sets. This also paves the way for other data structures like linked lists and heaps to be added to the list.



These features in other languages

Dot product is introduced in other language, such as Python, by importing Numpy library. The Haskell behavior of this feature act the same as the Python function, since it is a mathematical expression. Implementation is different since the feature uses the fix point operator to compute the result.

Set operations is introduced in many other languages, including Python which has Set built into it by default. Since the logic of set operations is the same for both languages, this feature acts the exact same as it would in Python. The Haskell implementation uses the built in “filter” function to help perform the set operations union, intersection, and difference. The other operations use a list and insert values into it like a set.



Limitations

Dot Product Operation:

Through using the test case implemented in the code, there should be no error/limitation when using the feature. The test case include basic correctness, type check, empty vector, non-vector input and mismatch length.

Set Operations:

The biggest limitation in the set operations is that it only supports the type Num. This means that it the sets cannot take in any other values like Booleans or a set inside of a set.